

Lucas Mercier

PhD Candidate in Computer Vision applied to 3D Human pose estimation · Seeking Postdoctoral Position for late 2026/2027

Laboratoire d'Innovation ouverte en technologie de la santé · ÉTS Montréal

 [GitHub](#)  [LinkedIn](#)  [Google scholar](#)  [ResearchGate](#)

Portfolio <https://www.lucasmercier.me/>

EDUCATION	École de technologie supérieure (ÉTS) - Montréal, Canada - Ph.D. in computer science - Developed acquisition software to synchronize RGB and MoCap systems; built inpainting pipeline using GANs to remove reflective markers from images; benchmarked and improved SOTA monocular 3D pose estimation models for clinical gait parameters. - Supervisors: Carlos Vázquez, Neila Mezghani, Thierry Cresson	oct 2021 - Present
	Institut supérieur d'électronique de Paris (ISEP) - Paris, France - Engineering school (equiv. M.Sc.) - Specialization: Data analysis and artificial intelligence (years 2 & 3) - End of track research project: Myelin sheath segmentation using UNet for sclerosis estimation	2016 - 2021
	Wrexham University - Wrexham, UK Exchange semester - Software and mobile development	Jan 2018 - May 2018
RESEARCH PROJECTS	Main topics: Computer vision, 3D Human pose estimation, Gait analysis, Machine learning, Image and video processing, Motion analysis, AI for healthcare	
	Related projects	
	Visiting PhD Student - Institut de Biomécanique Humaine Georges Charpak - Finetuned and benchmarked SOTA pose estimation model for gait parameter estimation on pathological populations - Improved these 2D-to-3D lifting models by integrating visual features from RGB frames using multi-scale deformable offset sampling	Jan 2026 - Mar 2026
	Real-time Markerless Pose Estimation for Movement Coaching - Flowtion (Co-founder) - Funded by Austria Wirtschaftsservice (55k€) - Led technical architecture and development as the primary engineer on a two-person engineering team - Built server-side pose estimation pipeline using WebRTC (aiortc) to stream client video for inference with pose models (HRNet, ViTPose etc.)	Sept 2024 - Present
	Pose Estimation Benchmarking Framework - Closed-source · Industry collaboration - Unified plugin-oriented framework for training and evaluating SOTA 2D/3D pose estimation models, designed for reproducibility and fair cross-model comparison. - Used across multiple PhD projects and an industry collaboration for clinical markerless motion analysis.	2023 - Present
	R&D Computer Vision Intern - Storelift - Implemented SOTA multi-object tracking algorithms (MOTNeuralSolver, FairMOT) for person tracking in autonomous retail environments. - Developed a cross-camera object re-identification pipeline using TorchReID. - Built a benchmarking framework to compare algorithm performance across metrics and camera configurations.	Feb 2021 - Aug 2021

PUBLICATIONS

JOURNALS

2026 Mercier L.,Cresson T. ,Gervais S. ,Mezghani N. ,Vázquez C. , Effect of markers in training dataset for markerless applications in biomechanics, Journal of Biomechanics [\[DOI\]](#)

CONFERENCES

2026 Mercier L.,Marois B. ,Fuentes A. ,Cagnin A. ,Cresson T. ,Vazquez, C. , Using knee kinesiography to train markerless pose estimation models for first-line kinematic assessment in knee osteoarthritis, OARSI World Congress on Osteoarthritis · April 2026 [\[DOI\]](#)

2025 Mercier L.,Cresson T. ,Mezghani N. ,Vázquez C. , Framework for the application of markerless motion capture to biomechanics, Multidisciplinary Biomechanics Journal · 2025 [\[DOI\]](#)

2025 Naaïm A. ,Rozaire J. ,Mercier L.,Begon M. ,Cherni Y. , Markerless motion capture accuracy in children with cerebral palsy and typically developing children, Multidisciplinary Biomechanics Journal · 2025 [\[DOI\]](#)

2025 Naaïm A. ,Rozaire J. ,Mercier L.,Duprey S. ,Begon M. , Evaluation of 3D marker-less motion capture precision in upper limb children movement, International Society of Biomechanics · Stockholm, Sweden · 2025

2025 Mercier L.,Cresson T. ,Mezghani N. ,Vázquez C. , Étude de l'impact de la présence des marqueurs pour l'estimation de la pose humaine, 92e Congrès de l'Acfas · Montréal, Canada · 2025

TEACHING
EXPERIENCE

Teaching assistant - ETS - Montreal

- GT411 - Digital Imaging (Fall 2024, Winter 2025, Fall 2025)

GRANTS &
AWARDS

Mobility Scholarship at Institut de Biomécanique Humaine Georges Charpak - Société de Biomécanique (International French-speaking Society of Biomechanics) 2025

Excellence grant for scientific communication activities - ÉTS Library 2024

SKILLS

Languages: Python, JS & TypeScript, SQL / Postgres, JAVA, C#

CV & ML: PyTorch, TensorFlow, OpenCV, scikit-learn

Tools: Git, Docker, Linux, SLURM & HPC (Compute Canada)

LAB
IMPLICATION

Authored lab software development quality procedures (reproducibility standards, Git workflows, onboarding documentation) as part of ISO certification maintenance · LIO-ÉTS 2023-2025